Quantum Mechanics from Rule 110: Hilbert Space, Hamiltonian, and Born Rule

Nova Spivack

Independent Research

2026

Preprint.

Archival version (Zenodo): https://doi.org/10.5281/zenodo.20319431
UGP series hub (Zenodo): https://doi.org/10.5281/zenodo.20168144

Abstract

We apply 't Hooft's cellular automaton interpretation of quantum mechanics to the GTE cellular automaton f_{MDL} on $\mathbb{Z}_7^5$, the 16,807-state visible-sector ring encoding the Standard Model generation structure.

The complete orbit decomposition of f_{MDL} (**A**) reveals a single cycle: the vacuum fixed point $(0, 0, 0, 0, 0)$ with $E = 0$. The physical Hilbert space is one-dimensional; f_{MDL} belongs to 't Hooft's information-loss regime. SM generation states are transients with tail lengths $\ell(\mathrm{gen}_1) = 3 > \ell(\mathrm{gen}_2) = 2 > \ell(\mathrm{gen}_3) = 1$, matching the empirical stability hierarchy (**A**). A direct eigenvalue-to-mass correspondence is falsified on every formulation tested (**A**).

The **Two-Role Structural Principle** (**A/D**) separates roles: the cogwheel construction provides Hilbert space structure, unitarity, and the Born rule ($\mathbf{A}_{\mathrm{Lean}}$, Lean-certified); the GTE N_{eff} cascade provides the mass spectrum.

The seven $\mathbb{Z}_7$ winding-equivalence classes of $\mathbb{Z}_7^5$ ($\mathbf{A}_{\mathrm{Lean}}$, `gauge_arithmetic_identification`) are identified as 't Hooft's information-equivalence gauge classes (**A/D**): the five SM winding sectors $\{0, 2, 3, 4, 6\}$ yield $U(1)_{\mathrm{EM}}$, $SU(2)_L$, and $SU(3)_c$; sector $W = 1$ predicts the SU(5) Y-leptoquark; the X-leptoquark ($Q = +4/3$) cohabits $W = 4$.

The AFCA causal graph is update-schedule-independent (Lean-certified); discrete Minkowski causal cones are embedded with coordinate surjection certified. Full Minkowski bijection remains open (**D**).

Contents

Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A/D} > \mathbf{A} > \mathbf{D}$$

$\mathbf{A}_{\text{Lean}}$: machine-checked exact arithmetic in Lean 4, zero `sorry`.
A/D: full analytic derivation; computationally confirmed.
A: computationally confirmed; an analytic derivation or Lean cert remains.
D: exploratory / long-term open.

1 Introduction

Whether quantum mechanics is fundamental or emergent from an underlying deterministic substrate is one of the deepest open questions in theoretical physics. 't Hooft's cellular automaton interpretation of quantum mechanics [15] provides a rigorous framework in which quantum states are superpositions of ontological basis states ("beables"), the Hamiltonian is defined by the periods of deterministic orbits, and the Born rule is a consequence of information loss at the coarse-graining scale. The framework accommodates both fully reversible automata (Chapter 2 of [15]) and irreversible automata with information loss (Chapter 7), and shows that the two regimes lead to qualitatively different quantum structures.

The UGP Physics programme, the quantitative branch of the Reflexive Reality programme [5,12] identifies Rule 110 as the minimal description-length (MDL) cellular automaton compatible with the Standard Model generation structure. The GTE cellular automaton f_{MDL} is a $\mathbb{Z}_7$-valued extension of Rule 110 over a five-cell ring $\mathbb{Z}_7^5$ (16,807 states), whose charge and winding assignments reproduce the SM particle spectrum [6]. The five-cell visible-sector ring has been used to derive the Weinberg angle [2], the CKM matrix structure [4], emergent spacetime dimension [7], and GUT charge quantization [6].

The present paper asks: what quantum mechanical structure does 't Hooft's framework assign to the f_{MDL} automaton? The answer turns out to be precisely the information-loss regime (Chapter 7): f_{MDL} on $\mathbb{Z}_7^5$ is a maximal information eraser — 98.71% of states are Garden-of-Eden states with no predecessor, and every trajectory reaches the vacuum fixed point in at most seven steps. The resulting physical Hilbert space is one-dimensional, and the SM generation states are transient excitations whose stability ordering is encoded in their tail lengths.

What this paper does and does not claim. We prove (at **A** or **A/D**) that:

- the f_{MDL} Hilbert space has dimension 1, spanned by $|\text{vac}\rangle$, with $E = 0$;
- the SM generation stability ordering is reproduced qualitatively by the tail-length hierarchy;
- the SM gauge group $G_{\text{SM}} = SU(3)_c \times SU(2)_L \times U(1)_{\text{EM}}$ can be derived from the $\mathbb{Z}_7$ winding-sector information-equivalence classes.

We do *not* claim to derive the Schrödinger equation from first principles (the continuum limit from a discrete CA to continuous quantum field theory is an open problem, **D**), nor do we claim that the cogwheel eigenvalues determine the SM mass spectrum (they are falsified as a mass formula at **A**). The gauge identification of Section 4 is 't Hooft's own conjecture (§9.3 of [15]) applied to the specific GTE automaton; its Lean 4 certification is identified as a target for future work.

The paper is organised as follows. Section 2 presents the f_{MDL} Hamiltonian spectrum and the Hilbert space structure, including the dark-sector contrast. Section 3 states and justifies the Two-Role

Structural Principle; this separation is forced by the failure of the direct cogwheel eigenvalue-to-mass correspondence (demonstrated in §3.1). Section 4 derives the SM gauge groups from $\mathbb{Z}_7$ information-equivalence classes. Section 5 discusses the Born rule and the role of measurement. Section 6 connects AFCA causal invariance to Lorentz structure and special-relativistic time dilation. Section 7 collects open problems.

2 The f_{MDL} Hilbert Space

't Hooft's information-loss regime (Chapter 7 of [15]) yields a low-dimensional physical Hilbert space; SM generations appear as transient excitations rather than energy eigenstates.

2.1 Cogwheel construction on $\mathbb{Z}_7^5$

In 't Hooft's cogwheel framework [15], an ontological CA with state space Ω defines a quantum system as follows. Let $T : \Omega \to \Omega$ be the deterministic evolution operator. The physical Hilbert space $\mathcal{H}_{\mathrm{phys}}$ is spanned by the *periodic orbit states*: if s belongs to a cycle of period N under T, then it contributes N energy eigenstates with eigenvalues $E_k = 2\pi k/N$, $k = 0, \ldots, N-1$, to $\mathcal{H}_{\mathrm{phys}}$. Transient states — those that eventually reach a cycle but are not themselves cycle states — do not contribute cycle eigenstates; they are the *Garden-of-Eden* (GoE) states if they have no predecessor under T.

Remark 2.1 (CA ontology and Wolfram Physics). The present construction grounds quantum mechanics in ontological CA states (beables) following 't Hooft's cogwheel mechanism; an independent programme grounding quantum mechanics in CA-like "rules" and identifying physical states with CA configurations is developed in [16] (§8.2). The two approaches share the CA-level ontological foundation but differ in their computational substrate and derivation strategy.

We apply this construction to $f_{\mathrm{MDL}} : \mathbb{Z}_7^5 \to \mathbb{Z}_7^5$, the canonical 14-entry lookup table (Lean-certified: `fmdl_nonzero_count_14` [5]). An exhaustive forward-orbit computation over all $7^5 =$ 16,807 states yields the following orbit decomposition (**A**).

Table 1: Complete orbit structure of f_{MDL} on $\mathbb{Z}_7^5$. All values are exact (exhaustive computation).

Quantity	Value	Cert.
Total states	16,807	exact
Distinct cycles	1	**A**
Cycle states	1 (vacuum only)	**A**
Transient states	16,806 (99.994%)	**A**
Garden-of-Eden states	16,590 (98.71%)	**A**
Maximum tail length	7 steps	**A**
Vacuum predecessor count	14,147	**A**

The unique cycle is the vacuum fixed point $(0,0,0,0,0)$, which has cycle length $N = 1$ and hence a single eigenvalue $E_0 = 0$. Standard Model generations appear as transients (gen$_1$ tail = 3 GoE steps, gen$_2$ tail = 2, gen$_3$ tail = 1), not as separate cycle sectors. On $\mathbb{Z}_7^4$ (dark-sector reduction), three period-2 cycles yield a five-dimensional physical Hilbert space $\{E=0\} \cup \{E=\pi\}$ doublets (`CUP3DUniqueness.lean`, `dark_sector_orbit_structure` in `GUTStructure.lean` §35, zero `sorry`).

Theorem 2.2 (Dimension of the f_{MDL} Hilbert space, **A**)**.** *The physical Hilbert space $\mathcal{H}_{\mathrm{phys}}$ of the f_{MDL} cogwheel on $\mathbb{Z}_7^5$ is one-dimensional:*

$$\mathcal{H}_{\mathrm{phys}} \;=\; \mathbb{C}\,|\mathrm{vac}\rangle, \qquad \hat{H}\,|\mathrm{vac}\rangle = 0.$$

The spectral gap is 2π (maximum possible), and $E = 0$ is the unique eigenvalue.

Proof. By exhaustive orbit decomposition (**A**, Table 1), the only cycle is the vacuum fixed point $(0,0,0,0,0)$ of period $N = 1$. Under 't Hooft's prescription, only cycle states contribute to $\mathcal{H}_{\mathrm{phys}}$; a period-1 cycle contributes one eigenvalue $E_0 = 2\pi \cdot 0/1 = 0$. No other cycles exist, so $\dim \mathcal{H}_{\mathrm{phys}} = 1$. □

Remark 2.3 (Quotient Hilbert space construction)*.* The passage from the full ontological Hilbert space to the physical Hilbert space follows 't Hooft's Chapter 7 quotient construction. Define the long-time equivalence relation: $s \sim s'$ if $\lim_{t\to\infty} f^t_{\mathrm{MDL}}(s) = \lim_{t\to\infty} f^t_{\mathrm{MDL}}(s')$, i.e. two CA states are identified if they share the same long-time attractor. The full ontological Hilbert space is $\mathcal{H}_{\mathrm{full}} = \mathrm{span}_{\mathbb{C}}\{|s\rangle : s \in \mathbb{Z}_7^5\}$ (dimension $7^5 = 16{,}807$). Since f_{MDL} on $\mathbb{Z}_7^5$ has a unique cycle — the vacuum fixed point $(0,0,0,0,0)$ — every trajectory converges to the vacuum. Consequently $\lim_{t\to\infty} f^t_{\mathrm{MDL}}(s) = (0,0,0,0,0)$ for all $s \in \mathbb{Z}_7^5$, and the equivalence relation collapses all 16,807 states into a single equivalence class. The physical Hilbert space $\mathcal{H}_{\mathrm{phys}} = \mathcal{H}_{\mathrm{full}}/\!\sim$ is therefore one-dimensional: $\mathcal{H}_{\mathrm{phys}} = \mathbb{C}|\mathrm{vac}\rangle$.

The SM generation states gen_1, gen_2, gen_3 are *transient excitations* in this quotient picture: they are not members of the equivalence class (they are not cycle states) but are pre-image chains leading to it, with tail lengths $\ell(\mathrm{gen}_1) = 3$, $\ell(\mathrm{gen}_2) = 2$, $\ell(\mathrm{gen}_3) = 1$ (Table 2).

A rough calibration of the CA timestep τ_{CA} follows from the muon (gen_2) tail length. The muon has tail length 2, so it decays to vacuum in 2 CA steps: $\tau_\mu \approx 2\tau_{\mathrm{CA}}$. Using the empirical muon lifetime $\tau_\mu = 2.197\,\mu$s, this gives $\tau_{\mathrm{CA}} \approx 1.1\,\mu$s as an effective CA timestep for second-generation decay. This is an order-of-magnitude estimate only (**D**: the CA step may correspond to many underlying fundamental steps; no quantitative mass derivation is claimed here). The figure $1.1\,\mu$s is a purely formal ratio; it does not imply the CA timestep is macroscopic. The physical CA timestep is expected to be of order the Planck time ($\sim 10^{-44}$ s), many orders of magnitude below the muon lifetime; the generation-tail ratio 2:1 is a dimensionless combinatorial count, not a physical time interval. The tail lengths 3:2:1 reproduce the qualitative stability ordering exactly, but the quantitative decay rates require a separate dynamical calculation beyond orbit combinatorics.

2.2 SM generation orbit: transient tail

The three SM generation representative states have the following orbit properties (**A**):

The SM generation orbit is thus a directed tail $\mathrm{gen}_3 \to \mathrm{gen}_2 \to \mathrm{gen}_1 \to \mathrm{vac}$ of total length 3 terminating at the vacuum. This places f_{MDL} firmly in 't Hooft's Chapter 7 (information-loss) regime rather than the reversible Chapter 2 regime.

Tail-length stability hierarchy (A). The tail lengths are ordered $\ell(\mathrm{gen}_1) = 3 > \ell(\mathrm{gen}_2) = 2 > \ell(\mathrm{gen}_3) = 1$, in exact correspondence with the empirical stability ordering $m(e) < m(\mu) < m(\tau)$ (lightest and most stable for gen_1, heaviest and most unstable for gen_3). Moreover, gen_1 is a Garden-of-Eden state (zero predecessors): the vacuum dynamics can never *create* the first-generation state from below, providing a CA-level account of electron stability.

This qualitative ordering is a **A** result. The tail lengths 3:2:1 do not predict quantitative mass ratios — the actual lepton masses are 1:207:3477, far from the linear sequence $3:2:1$ in either direction — and we make no such claim.

Table 2: SM generation states in the f_{MDL} orbit on $\mathbb{Z}_7^5$.

State	Tuple	On cycle?	Tail length	Predecessors	Physical analog
gen_1	$(1,5,2,2,1)$	No (transient)	3	0 (GoE)	e,u,d
gen_2	$(2,5,2,0,2)$	No (transient)	2	1	μ,c,s
gen_3	$(5,6,5,3,5)$	No (transient)	1	1	τ,t,b
vacuum	$(0,0,0,0,0)$	Yes (cycle)	0	14,147	vacuum

2.3 Dark sector: contrasting Hilbert space

The GTE dark sector ring $\mathbb{Z}_7^4$ (2,401 states) has a structurally different cycle decomposition, confirmed at $\mathbf{A}_{\mathrm{Lean}}$ by Lean 4 exhaustive proof (theorems `dark_sector_orbit_structure`, `dark_sector_period2_exhaustive`):

Table 3: Comparison of visible-sector $\mathbb{Z}_7^5$ and dark-sector $\mathbb{Z}_7^4$ orbit structures.

Quantity	Visible ($\mathbb{Z}_7^5$)	Dark ($\mathbb{Z}_7^4$)
Distinct cycles	1	3
Non-vacuum cycles	0	2 (both period-2)
Hilbert space dimension	1	5
Energy eigenvalues	$\{0\}$	$\{0,\pi,\pi\}$
GoE fraction	98.71%	97.08%
Maximum tail length	7	4

The four dark cycle states are exactly the period-2 orbits of Rule 110 on the 4-cell binary ring (Lean-certified: `dark_sector_period2_exhaustive`). Their 't Hooft energy eigenvalues are $E = 0$ (vacuum, one state) and $E = \pi$ (two doublet pairs). The 5-dimensional dark Hilbert space with stable doublet states at $E = \pi$ provides a CA-level mechanism for dark-sector stability, in contrast to the purely transient visible sector. This is a new prediction of the GTE framework ($\mathbf{A}_{\mathrm{Lean}}$).

3 Two Irreducible Roles: A Structural Principle

3.1 The direct eigenvalue-mass correspondence fails

A natural conjecture is that the SM generation masses are proportional to the cogwheel Hamiltonian eigenvalues. We test this conjecture systematically.

T=3 cyclic approximation. Treating the SM generation orbit as a cyclic cogwheel of period $T = 3$ yields equally-spaced eigenvalues $E_k = 2\pi k/3$ for $k = 1, 2, 3$. The ratio spread between $E_k/(N_{\mathrm{eff}}(\mathrm{gen}_k)/c_H)$ is 69.3%, far from the 0% that proportionality would require.

T=4 cyclic approximation. Including the vacuum as a fourth state and assigning $T = 4$ gives a ratio spread of 63.0% — also far from proportionality.

Direct mass comparison. Any period-T cogwheel predicts mass ratios $1 : 2 : \cdots : T$ (equally spaced). The PDG lepton masses give $m_e : m_\mu : m_\tau \approx 1 : 207 : 3477$, a discrepancy of 10,238% for m_μ/m_e and 115,807% for m_τ/m_e relative to the cogwheel prediction.

Alternative hypotheses. Orbit depth as an energy proxy fails because the depth ordering is *inverted* relative to the mass ordering (deeper = lighter, not heavier). The predecessor-count hypothesis fails because there is only one eigenvalue $E = 0$ in $\mathbb{Z}_7^5$, leaving no distinct eigenvalues to carry degeneracy multiplicities. All N_{eff} values (73, 42, 275) differ from the predecessor counts (0, 1, 1) for $\text{gen}_{1,2,3}$.

Proposition 3.1 (Eigenvalue-mass falsification, **A**)**.** *The direct correspondence $E_k \propto m(gen_k)$ fails on every formulation: equally-spaced cogwheel ($>$ 10,000% discrepancy), orbit-depth proxy (ordering inverted), degeneracy assignment (only $E = 0$ exists). The near-miss $|$tail-3 state count$| = 75$ vs. $N_{\text{eff}}(gen_1) = 73$ (2.7% difference) is noted but does not constitute a theorem.*

3.2 Statement of the Two-Role Structural Principle

The failure of the direct eigenvalue-mass correspondence points to a fundamental separation of roles within the GTE framework.

Remark 3.2 (Two-Role Structural Principle, **A/D**). In the GTE framework, the 't Hooft cogwheel construction and the N_{eff} arithmetic of the Generative Triple Evolution (GTE) cascade play distinct, complementary roles:

1. **Cogwheel role (QM structure):** The f_{MDL} cogwheel on $\mathbb{Z}_7^5$ provides the physical Hilbert space $\mathcal{H}_{\text{phys}} = \mathbb{C}\,|\text{vac}\rangle$, the Hamiltonian $\hat{H}$ with unique vacuum eigenvalue $E = 0$, the unitary evolution operator, the Born rule (via the information-loss mechanism of Section 5), and the tail-length stability ordering of SM generations.

2. **GTE arithmetic role (mass content):** The N_{eff} cascade provides the SM generation mass scale via the beable superposition formalism: $m(\text{gen}_k) \propto N_{\text{eff}}(\text{gen}_k)/c_H$, with $N_{\text{eff}}(\text{gen}_1) = 73$, $N_{\text{eff}}(\text{gen}_2) = 42$, $N_{\text{eff}}(\text{gen}_3) = 275$. These numerical values are inputs to the cogwheel, not outputs from it.

These contributions are non-redundant and complementary: the cogwheel provides the quantum mechanical scaffolding; the GTE cascade provides the content. This is an organizing structural principle (**A/D**), not a machine-certified theorem: the arithmetic skeleton (orbit decomposition, Hilbert space dimension, tail lengths, N_{eff} values) is Lean 4-certified at zero `sorry`, but the physical identification of each role with quantum-mechanical structure and mass content respectively is a conditional derivation that awaits formal Lean 4 certification of the inter-layer mapping. `TwoRoleTheorem.lean` (zero `sorry`) certifies `two_role_structural_separation`, `cogwheel_equal_spacing_fails`, and `two_roles_non_redundant`: equal-spacing cogwheel eigenvalues cannot reproduce GTE mass ratios.

3.3 Physical significance

The Two-Role Structural Principle explains why the direct eigenvalue-mass correspondence fails: masses are determined not by the period of the cycle in $\mathbb{Z}_7^5$ (which is trivially 1) but by the arithmetic of the GTE cascade that generates the Z_7 state alphabet. The cogwheel is the *kinematic* framework; the GTE arithmetic is the *dynamical* input that fills in the mass spectrum.

This separation parallels the structure of standard quantum field theory, where the Hilbert space and the canonical commutation relations (the kinematic framework) are universal, while the specific mass spectrum depends on the Lagrangian (the dynamical input). In the present framework, the Lagrangian's role is played by the GTE N_{eff} cascade.

4 Information-Equivalence Classes and Gauge Structure

4.1 Information erasure and equivalence classes

As established in Section 2, f_{MDL} on $\mathbb{Z}_7^5$ erases all information within at most 7 steps: $T^7(s) = (0,0,0,0,0)$ for every $s \in \mathbb{Z}_7^5$. The naive forward-trajectory equivalence $s \sim s'$ iff $T^t(s) = T^t(s')$ for some $t \geq 0$ therefore yields only a single equivalence class (all 16,807 states are equivalent). This extreme form of information erasure produces no non-trivial gauge structure at the CA beable level.

To recover gauge structure, we follow 't Hooft's §9.3 prescription: the physically meaningful equivalence classes are defined by *observable quantum numbers* — the quantities that macro-scale observers can measure — not by the full CA configuration. For the GTE framework, the relevant observable quantum number is the $\mathbb{Z}_7$ winding number $W = \sum_{i=1}^5 w_i \bmod 7$.

4.2 Winding-sector equivalence classes

Definition 4.1 (Winding equivalence). *Two configurations $s, s' \in \mathbb{Z}_7^5$ are winding-equivalent, $s \sim_W s'$, if $\sum_i s_i \equiv \sum_i s'_i \pmod 7$.*

Proposition 4.2 (Winding class structure, $\mathbf{A}_{\text{Lean}}$). *There are exactly 7 winding-equivalence classes, each of size $7^4 = 2{,}401$ states, corresponding to $W \in \{0,1,2,3,4,5,6\}$. The SM sectors are $\{0,2,3,4,6\}$ (cardinality 5); the GUT mediator sectors are $\{1,5\}$ (cardinality 2).*

Proof. The map $\phi : \mathbb{Z}_7^5 \to \mathbb{Z}_7$, $\phi(s) = \sum_i s_i \bmod 7$, is a surjective $\mathbb{Z}_7$-module homomorphism. Its kernel has order $|\ker \phi| = 7^5/7 = 7^4 = 2{,}401$, so each fibre has exactly 2,401 elements. □

Remark 4.3 ($\mathbf{A}_{\text{Lean}}$ certification of partition arithmetic). The partition arithmetic — 7 uniform sectors of $7^4 = 2{,}401$ states each, SM sectors $\{0,2,3,4,6\}$ of cardinality 5, GUT sectors $\{1,5\}$ of cardinality 2, and Nat.Prime 7 (the algebraic guarantee of uniform sector sizes via Lagrange's theorem) — is machine-certified in Lean 4 (`gauge_arithmetic_identification`, `GUTStructure.lean` §55, zero sorry). Supporting theorems ($\mathbf{A}_{\text{Lean}}$, same file): `z7_sector_sizes`, `z7_sector_count`, `sm_sector_count`, `gut_leptoquark_sectors`, `gauge_sector_identification`. The physical identification (gauge transformation = info-equivalence class symmetry in the 't Hooft §9.3 sense) remains a **A/D** derivation.

The SM particle sectors and the remaining winding sectors are identified by their winding class ($\mathbf{A}_{\text{Lean}}$, Lean theorems `sm_sector_count` and `gauge_arithmetic_identification` in `GUTStructure.lean` §55):

The five SM winding sectors $\{0,2,3,4,6\}$ have cardinality 5, equal to the dimension of the fundamental representation $\bar{\mathbf{5}}$ of $SU(5)$. The two missing sectors $\{1,5\}$ correspond to the SU(5) Y-leptoquark mediator ($W = 1$, $Q = +\frac{1}{3}$) and the $\bar{u}$ (anti-up) sector ($W = 5$, $Q = -\frac{2}{3}$). The X-leptoquark ($Q = +\frac{4}{3}$, $3Q \equiv 4 \pmod 7$) cohabits winding sector $W = 4$ with the electron. The complete 7-sector structure thus predicts the $SU(5)$ GUT enlargement of the SM gauge group.

4.3 $\mathbf{U(1)_{EM}}$ from winding-sector phase freedom

Within winding sector $W = k$, the physical quantum states are superpositions

$$|\psi_k\rangle = \sum_{\{s\,:\,W(s)=k\}} \alpha_s\,|s\rangle.$$

Table 4: $\mathbb{Z}_7$ winding sectors and SM/GUT assignments.

Class W	States	SM assignment	Source
0	2,401	vacuum / neutrino sector	$\mathbf{A}_{\text{Lean}}$
2	2,401	up-quark sector	$\mathbf{A}_{\text{Lean}}$
3	2,401	W^+ boson sector	$\mathbf{A}_{\text{Lean}}$
4	2,401	electron / W^- sector; X-leptoquark ($Q = +\frac{4}{3}$) cohabits	$\mathbf{A}_{\text{Lean}}$
6	2,401	down-quark sector	$\mathbf{A}_{\text{Lean}}$
1	2,401	SU(5) Y-leptoquark mediator	**A/D**
5	2,401	$\bar{u}$ (anti-up quark, $Q = -\frac{2}{3}$)	$\mathbf{A}_{\text{Lean}}$

The gauge freedom within sector k is the freedom to multiply all amplitudes by a common phase: $|\psi_k\rangle \to e^{i\varphi}|\psi_k\rangle$. This is a $U(1)$ symmetry.

For the electron sector $W = 4$ (charge $Q = -1$), the electromagnetic gauge transformation is $|e^-\rangle \mapsto e^{iQ\theta}|e^-\rangle$. The connection between the $\mathbb{Z}_7$ winding number and the electric charge Q is established at $\mathbf{A}_{\text{Lean}}$ by the assignment `winding_class_sm_assignment` [6]: $W = 0 \to Q = 0$, $W = 2 \to Q = +2/3$, $W = 3 \to Q = +1$, $W = 4 \to Q = -1$, $W = 6 \to Q = -1/3$.

Result: $U(1)_{\text{EM}}$ arises from the phase freedom within each $\mathbb{Z}_7$ winding-equivalence class (**A/D** identification, $\mathbf{A}_{\text{Lean}}$ for the $\mathbb{Z}_7 \to Q$ map).

4.4 $\mathbf{SU(3)_c}$ from the multiplicative $\mathbf{Z_3}$ subgroup

The multiplicative group $\mathbb{Z}_7^*$ has order 6, and its unique subgroup of order 3 is $Z_3 = \{1, 2, 4\}$ (generated by 2, since $2^3 \equiv 1 \pmod 7$). This is Lean-certified ($\mathbf{A}_{\text{Lean}}$): `z7_color_subgroup_closed`, `z7_color_subgroup_generator`, `w_u_in_color_subgroup`.

The up-quark sector $W = 2$ satisfies $2 \in Z_3$. The three colour states of the up quark correspond to multiplication of the base winding by $\{1, 2, 4\}$: $w_u \cdot \{1, 2, 4\} = \{2, 4, 1\} \pmod 7$. This defines a faithful 3-dimensional permutation representation of Z_3 on the three colour states, which embeds into (and generates the centre of) $SU(3)_c$.

Result: $SU(3)_c$ arises from the $Z_3 = \{1, 2, 4\} \subset \mathbb{Z}_7^*$ action on the up-quark winding sector (**A/D** identification, $\mathbf{A}_{\text{Lean}}$ for the Z_3 arithmetic).

4.5 $\mathbf{SU(2)_L}$ from the weak-isospin doublet

The W^+ sector ($W = 3$) and the electron/W^- sector ($W = 4$) form a weak-isospin doublet with $T_3 = +1$ and $T_3 = -1$ respectively. The charged-current interaction $W^+ + d \to u$ corresponds to $W(3) + W(6) = 9 \equiv 2 \pmod 7 = W(u)$, i.e., a winding-conserving process ($\mathbf{A}_{\text{Lean}}$, `su2l_charge_assignment_z7_discriminator`).

The gauge freedom within the doublet $\{W = 3, W = 4\}$ is the group of 2×2 unitary transformations that preserve the doublet structure, which is $SU(2)_L$.

Result: $SU(2)_L$ arises from the $\{W = 3, W = 4\}$ doublet winding pair (**A/D** identification, $\mathbf{A}_{\text{Lean}}$ for the doublet arithmetic).

Remark 4.4 (Doublet structure machine-certified at $\mathbb{Z}_7$ layer ($\mathbf{A}_{\text{Lean}}$), zero `sorry`)*.* The weak isospin doublet partition and charged-current conservation are machine-certified at the species-winding layer. The $\mathbb{Z}_7$ species formula $W_B = 4k \bmod 7$ forces $\Delta W_B = 4$ between doublet partners for both the lepton doublet ($\nu \leftrightarrow e^-$) and the quark doublet ($u \leftrightarrow d$); W_B conservation holds at all four SM charged-current vertices; and the $W^\pm$ winding numbers $W_B(W^+) = 3$,

$W_B(W^-) = 4$ are uniquely determined by the conservation constraint. Lean-certified in `weak_isospin_identification` (`WeakIsospin.lean`, zero `sorry`, $\mathbf{A}_{\text{Lean}}$). The $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ substrate automorphism group contains no $\text{SU}(2)_L$ factor ($\mathbf{A}_{\text{Lean}}$); the doublet structure is a consequence of $\mathbb{Z}_7$ winding arithmetic rather than a gauge symmetry postulate. The V–A chiral structure is $\mathbf{A}_{\text{Lean}}$: `ChiralPairVA.lean` (zero `sorry`) proves `fmdl_va_bundle`—the two-layer chiral CA (Rule 110 + Rule 124) produces a ratio of 32/125 L↔R mismatches, with $32/125 \neq 1$ (parity not conserved), $0 < 32/125 < 1$ (left-handed preference), and asymmetry degree 93/125. Every mismatch involves W^+ ($W_B = 3$) as a neighbor, connecting V–A chirality to the CC winding structure of `weak_isospin_identification`. The identification of 32/125 with the SM V–A coupling constant remains **A/D**.

Remark 4.5 (W^- as CPT-adjoint shadow in $\mathcal{H}$ (**A/D**)). The W^- boson does not appear as a CA beable: f_{MDL} produces winding sector $W = 3$ (W^+) but never $W = 4$ under the GTE map (`fmdl_never_outputs_4`; zero sorry [5]). Yet W^- is forced to appear as an operator in the Hilbert space $\mathcal{H} = \text{span}_{\mathbb{C}}\{|\text{CA states}\rangle\}$ by two independent constraints:

- *Unitarity*: the W^- creation operator $a^\dagger_{W^-} = (a_{W^+})^\dagger$ is the Hermitian adjoint of the W^+ annihilation operator; any unitary Hilbert space must contain both.
- *CPT*: any Lorentz-invariant local QFT limit inherits CPT invariance (Lüders–Pauli theorem), forcing W^--mediated processes wherever W^+-mediated processes occur.

The $\mathbb{Z}_7$ additive structure reinforces this: $w(W^-) = -w(W^+) \bmod 7 = 4$ is the unique algebraic inverse of $w(W^+) = 3$, and both weak vertices $u \to d + W^+$ and $d \to u + W^-$ are winding-conserving mod 7. The full $\text{SU}(2)_L$ isospin algebra (T_+, T_-, T_3) is therefore realised at the level of $\mathbb{Z}_7$ winding shifts. The confidence is **A/D**: the Hilbert-space and $\mathbb{Z}_7$ arithmetic steps are $\mathbf{A}_{\text{Lean}}$, while the CPT premise requires the CA→QFT embedding to produce a Lorentz-invariant local QFT (open premise, **A/D** per 't Hooft's programme §9.3).

4.6 Hypercharge and Weinberg angle ($\mathbf{A}_{\text{Lean}}$)

The winding-sector assignments give the correct electromagnetic charges ($\mathbf{A}_{\text{Lean}}$). The extension to the full electroweak hypercharge $U(1)_Y$ via $Y = 2(Q - T_3)$ is machine-certified: for every member of the quark and lepton $\text{SU}(2)_L$ doublets (u-quark, d-quark, electron, neutrino) the GTE charge assignments satisfy $Y = 1/3$ (quark doublet) and $Y = -1$ (lepton doublet), certified by `quark_doublet_hypercharge` and `lepton_doublet_hypercharge` (§49, $\mathbf{A}_{\text{Lean}}$). The Weinberg-angle formula $\sin^2\theta_W = N_{\text{gen}}/c_H = 3/13$ (established at **A** [2]) is machine-certified in this framework (§73 `HyperchargeConsistency`, $\mathbf{A}_{\text{Lean}}$, `ugp-lean`, zero sorry).

4.7 Summary of the gauge derivation

Theorem 4.6 (SM gauge group from winding equivalence, **A/D**)**.** *The SM gauge group* $G_{\text{SM}} = SU(3)_c \times SU(2)_L \times U(1)_{\text{EM}}$ *is derived from the* $\mathbb{Z}_7$ *winding-sector information-equivalence classes of* f_{MDL} *on* $\mathbb{Z}_7^5$ *via 't Hooft's §9.3 identification:*

- $U(1)_{\text{EM}}$*: phase freedom within each winding sector;* $\mathbb{Z}_7 \to Q$ *map* $\boldsymbol{A}_{\text{Lean}}$*.*
- $SU(2)_L$*:* $\{W = 3, W = 4\}$ *doublet pair; doublet arithmetic* $\boldsymbol{A}_{\text{Lean}}$*.*
- $SU(3)_c$*:* $Z_3 = \{1, 2, 4\} \subset \mathbb{Z}_7^*$ *acting on winding-2 sector;* Z_3 *arithmetic* $\boldsymbol{A}_{\text{Lean}}$*.*

- *$U(1)_Y$: hypercharge $Y = 2(Q - T_3)$ consistent for all SM doublet members ($\boldsymbol{A}_{\text{Lean}}$, `quark_doublet_hypercharge`, `lepton_doublet_hypercharge`); Weinberg angle $\sin^2\theta_W = N_{\text{gen}}/c_H = 3/13$ machine-certified (`HyperchargeConsistency`, §73, $\boldsymbol{A}_{\text{Lean}}$).*

The overall confidence is **A/D**: each arithmetic component is $\mathbf{A}_{\text{Lean}}$ (Lean-certified), while the identification of $\mathbb{Z}_7$ winding classes as 't Hooft's information-equivalence gauge classes remains at the level of 't Hooft's own §9.3 conjecture. Formal Lean 4 certification of this identification remains open.

Remark 4.7 (Partial SM gauge group Lean bundle certificate). The three $\mathbb{Z}_7$ channel identifications are packaged in `sm_gauge_group_certificate` (`ugp-lean/UgpLean/Algebra/SMGaugeGroup.lean`, zero sorry): (i) $\mathrm{SU}(3)_c$ via $F_{21} \hookrightarrow \mathrm{SU}(3)$ embedding and gluon count ($\mathbf{A}_{\text{Lean}}$ arithmetic; Burnside named axiom `f21_burnside_full_enveloping_algebra`, **A/D**); (ii) $\mathrm{U}(1)_Y$ via `gte_charge_formula` and Weinberg angle ($\mathbf{A}_{\text{Lean}}$); (iii) $\mathrm{SU}(2)_L$ via channel count and MDL gauging (`su2l_l2_from_phimdl_potential_catad`, $\mathbf{A}_{\text{Lean}}$, zero named axioms). Bundle theorems `sm_gauge_group_three_factors` and `sm_gauge_factors_independent` are zero sorry. Full certification — the formal Lie-algebra product $\mathrm{SU}(3) \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ — requires discharge of the Burnside named axiom and is not yet complete.

CA-level gauge non-invariance. It is important to note that f_{MDL} itself is *not* locally $\mathbb{Z}_7$-invariant: a local shift on a single cell generically changes the image under f_{MDL}. This is entirely consistent with the Two-Role Structural Principle (Theorem 3.2): gauge invariance is a property of the *quantum superposition layer*, not the CA beable layer. The CA dynamics fix the Hilbert space and the Born rule; gauge invariance is an emergent symmetry of the quantum states built from winding-equivalence classes.

5 Information Loss, Born Rule, and Measurement

5.1 Born rule from information loss

't Hooft's derivation of the Born rule (Chapters 4 and 7 of [15]) proceeds in two steps. First, the unitary evolution operator $U(t)$ on $\mathcal{H}_{\text{phys}}$ is constructed as the exponential of the cycle-Hamiltonian $\hat{H}$. Second, when the physical system undergoes information loss (transitions to the vacuum attractor), the relative probabilities of outcomes are determined by the squared moduli of the quantum amplitudes in the Born-rule sense.

For f_{MDL} on $\mathbb{Z}_7^5$, the information-loss mechanism is maximal: every state converges to the vacuum in at most 7 steps. A measurement performed on the system at macro-scales corresponds to asking which winding sector W the CA configuration belongs to, and the probability of obtaining outcome $W = k$ is $P(W = k) = |\alpha_k|^2$ where α_k is the amplitude in the k-th winding-sector component of the quantum state. This is the Born rule.

The derivation is structural (**A/D**): it follows from the general framework of [15] applied to the specific f_{MDL} information-loss structure. An independent derivation of the Born rule in the UGP framework appears in an earlier paper in this series [3].

An independent derivation of the same principle appears in Wolfram's multiway graph framework [16] (§8.16): observer-induced decoherence is identified with the inaccessibility of alternative branches in the multiway causal graph, making quantum probability a consequence of the irreversibility of computational dynamics. The GTE derivation and the Wolfram derivation converge on the same physical conclusion — irreversibility of CA-level dynamics grounds quantum probability — through different mathematical formalisms (attractor convergence vs. multiway inaccessibility).

5.2 Comparison with the UGP Born rule derivation

An independent Born rule derivation was established in the UGP series [3, 11] via the information geometry of the Zone L1 and Zone L2 distinction (Zones L1/L2 are the two arithmetic classification levels established in P34 [8]: Zone L1 covers Turing-decidable / unitary-evolution questions; Zone L2 covers Turing-undecidable / measurement questions). NEMS Paper 13 (*Born Rule as a Closure Fixed Point*) proves that the Born rule is the unique quantum probability assignment in perfectly self-contained (PSC) theories, using the Busch–Gleason representation theorem; the derivation is Lean 4 formalized in the `nems-lean` library (`NemS.Quantum.BornFromPSC`, theorems `born_rule_from_psc` and `born_rule_projectors_from_psc`). In that framework, measurement corresponds to the transition into the undecidable diagonal of the computational substrate, which is identified with the transputation zone of the Rule 110 CA.

The two derivations are complementary:

- The 't Hooft derivation (**A/D**, present paper) is structural: it follows from the general framework of ontological CA and the information-loss mechanism.
- The Zone framework derivation is constructive: it assigns measurement to a specific computational zone of the Rule 110 CA.

Both derivations identify the Born rule with the irreversibility of the CA dynamics — 't Hooft via Chapter 7 information loss, the Zone framework via transition to the undecidable diagonal. The 't Hooft ontological–template lift (information loss on f_{MDL}) and NEMS Paper 13 (PSC closure $\Rightarrow$ noncontextual POVM-additive effect algebra $\Rightarrow$ Born rule via Busch–Gleason) are *not* the same theorem: they rest on different premises but converge on $P = |\psi|^2$. They are complementary, not in tension with the GTE substrate.

A further independent derivation, at a stronger certification level, is established in P41 and P42 [9, 13] via $\mathbb{Z}_7$-KG kink superselection (the Coleman–Mandelstam–Rajaraman topological argument). This route yields `born_rule_unconditional` (`BeableWindingPartitionInstance.lean`, zero `sorry`, zero custom physics axioms, $\mathbf{A}_{\mathrm{Lean}}$), with the Born probability $P(k) = |c_k|^2$ following unconditionally from $\mathbb{Z}_7$ superselection alone. This derivation is independent of the 't Hooft gauge-identification step (which remains **A/D** in the present paper) and provides the currently strongest certification of the Born rule in the UGP framework. Zone L1 (vacuum attractor / information loss) is structurally parallel to Zone L2 (transputation diagonal). A formal bridge theorem (PSC $\Rightarrow$ effect algebra from f_{MDL} dynamics alone) remains open; the constructive route via kink Fock lift (below) closes the representation step conditionally ($\mathbf{A}_{\mathrm{Lean}}$).

A third independent derivation appears in [1] (§8.15): the Born rule probability $|\psi|^2$ is identified with the relative weight of branches in the multiway graph, arising from the structure of the multiway causal graph over the computational state space. Whether this branch-weight derivation is formally equivalent to the PSC entropy derivation or the 't Hooft information-loss derivation is an open question.

A fourth independent derivation is established in P45 [14] via the Page–Wootters (PW) mechanism with the τ_c outer clock: the Born rule $P(k|\tau) = |\langle k|U(\tau)|\psi_0\rangle|^2$ is derived from the timeless universe state $|\Psi\rangle = \sum_\tau c_\tau|\tau\rangle \otimes U(\tau)|\psi_0\rangle$. The τ_c clock satisfies all PW prerequisites (non-degenerate Hamiltonian $H_{\tau_c} = \omega_c \sum_\tau \tau|\tau\rangle\langle\tau|$; orthogonal clock states; Wheeler-DeWitt constraint; **A/D**). The derivation is independent of the clock state distribution $\{c_\tau\}$: uniform (classical) or Gaussian (quantum) clock weights give identical conditional Born probabilities $P(k|\tau)$. At the discrete Level 1, the $\mathbb{Z}_7$ Born rule assigns probability $P(j|w) = 1/7$ to each position j for a winding eigenstate $|w\rangle$; non-uniform distributions arise from superpositions. The continuous Born rule $P(x) = |\psi(x)|^2$ is a Level 2 property supplied by Φ_{MDL} (P42).

5.3 Measurement and Zone L2

The non-reversible f_{MDL} dynamics (98.71% GoE states) place the GTE CA firmly in the irreversibility regime. In the zone framework, the undecidable diagonal of Rule 110 defines Zone L2, which is identified with the measurement domain. The transition of a transient SM state ($\text{gen}_{1,2,3}$) to the vacuum under f_{MDL} evolution is then the CA-level description of a measurement event: the winding sector W of the initial transient state becomes the classical measurement outcome as the system relaxes to the vacuum attractor.

This identification connects the 't Hooft Born rule (Section 5.1) with the Zone L2 framework at a structural level. The formal discrete measurement mechanism — characterizing f_{MDL} as the PSC-correction of p and proving the Three-Level MDL Unification — is developed in [10]. The precise equivalence theorem is a target for future work.

Remark 5.1 (Structural parallel with Wolfram branch foliation)*.* The identification of Zone L2 with the measurement domain is structurally parallel to the treatment of quantum measurement in [16] (§8.14), where measurement corresponds to the selection of a branch foliation in the multiway causal graph: the observer's branch becomes classical while information in alternative branches becomes inaccessible. In both frameworks, the quantum-to-classical transition is identified with an irreversible selection process in the underlying computational substrate. The physical mechanism differs: Zone L2 is defined by the undecidable diagonal of Rule 110; the Wolfram foliation is a spacelike slice of the multiway causal graph. A formal map between the two constructions is an open problem (see Section 7.6).

Remark 5.2 (Measurement problem in the GTE-CA interpretation; **A**)*.* The measurement problem asks how definite, classical outcomes emerge from unitary quantum evolution. In the GTE-CA interpretation, measurement outcomes are definite because Zone L2 adjudication by the $[D]$ coherence class is a non-computable but determinate selection among the transient beable states: the CA dynamics evolve a superposition of $\mathbb{Z}_7^5$ beable states to the Zone L2 boundary, and the PSC-forced $[D]$-selection irreversibly picks one outcome. This differs structurally from the many-worlds interpretation (no collapse; all branches realized) and from Copenhagen (collapse without mechanism); it also goes beyond decoherence approaches, which identify the preferred basis but leave the single-outcome selection unresolved. The GTE mechanism is closest to objective-collapse theories, with the collapse mechanism being the $[D]$-selection at the Zone L2 boundary rather than a postulated stochastic process. The preferred basis is fixed by the $\mathbb{Z}_7^5$ CA dynamics, as described in Remark 5.3: the $\mathbb{Z}_7^5$ winding-sector states are the unique ontological basis under which f_{MDL} acts as a deterministic map. The measurement problem in GTE therefore reduces to the Zone L2 adjudication question, whose mechanism is specified (PSC-forced $[D]$-selection) but whose complete formalization is linked to Conjectures C3 (TPC Completeness) and C4 (Lawvere–Physical Correspondence) of P34 [8].

Remark 5.3 (Preferred-basis selection by f_{MDL})*.* The preferred-basis problem — why quantum measurement outcomes appear in the particle/occupation-number basis rather than in an arbitrary unitary rotation of it — receives a structural resolution in the CA framework. In 't Hooft's ontological formulation [15], the *beable basis* is selected by requiring the CA evolution operator T to act as a permutation matrix on it: $T|s\rangle = |f_{\text{MDL}}(s)\rangle$ for every basis vector $|s\rangle$ with $s \in \mathbb{Z}_7^5$. This holds for the standard computational basis $\{|s\rangle : s \in \mathbb{Z}_7^5\}$ by construction. Any unitary rotation U of the Hilbert space replaces $|s\rangle$ by a superposition $U|s\rangle = \sum_{s'} U_{s's}|s'\rangle$, so T no longer acts as a permutation on the rotated basis: the rotated-basis evolution mixes many basis states into a single output, destroying the deterministic single-valued structure. The preferred basis is therefore selected uniquely by the criterion that f_{MDL} acts as a deterministic function (a permutation-like map, technically a non-injective endomorphism in the information-loss case) on it. No other basis satisfies this criterion. This matches 't Hooft's own preferred-basis argument ([15], §2.3): the ontological basis is the unique one in which the time-evolution operator is diagonal and classical. In the GTE context, this means the $\mathbb{Z}_7^5$ winding-sector states are the ontological beables, and quantum superpositions of winding sectors are informational (superpositions of our knowledge of the CA state), not ontological coexistences of distinct CA configurations. The Zone L2 adjudication then selects the specific winding-sector outcome during a measurement event, consistent with the decoherence programme selecting the preferred beable basis in open-system evolution.

5.4 MDL structural layer and kink Fock lift

`BornRuleMDL.lean` (zero `sorry`) certifies that MDL-minimal completion selects f_{MDL} (`mdl_selects_minimum_description`) and that inverse-complexity branch weights normalize to a probability measure (`born_weight_from_mdl_minimality`). Conditional on a constructive `BeableAmplitudeHypothesis`, `born_rule_from_psc_mdl` gives sector Born weights that partition unity, compatible with the winding-sector EffectMeasure schema (below).

`FockSpaceKink.lean` supplies the amplitude hypothesis constructively: four GTE kink Fock modes with certified q_Φ labels; `kinkCreation`/`kinkAnnihilation` on occupation-number states; `BeableWindingPartition` uniform sector lift; `fock_beable_amplitude_normalized` and `born_rule_from_fock_lift` connect to the MDL layer. Open gaps: canonical $\mathbb{Z}_7$-KG soliton quantization (Jackiw–Rebbi limit); identification of MDL weight with $|\varphi_k|^2$; explicit beable-index partition for every physical superposition.

Remark 5.4 (Fock-space structure and CA$\leftrightarrow$QFT embedding). The CMCA state space admits a Fock-space decomposition via the MDL mode expansion (**A/D**, conditional): the Hilbert space $\mathcal{H}_{\mathrm{MDL}}$ is spanned by occupation-number states $|n_1, n_2, \ldots\rangle$ for $\mathbb{Z}_7$ normal modes via the GNS construction from the MDL vacuum state ω_{vac} and inductive limit over finite sub-lattices. The 't Hooft CA$\leftrightarrow$QFT embedding map $\Phi : \mathcal{H}_{\mathrm{CMCA}} \to \mathcal{H}_{\mathrm{QFT}}$ decomposes as $\Phi =$ (Hilbert-space map, here) $\circ$ (`cmca_continuum_limit_is_phimdl`, $\mathbf{A}_{\mathrm{Lean}}$ conditional): the structural reduction is established; full unconditional proof awaits canonical $\mathbb{Z}_7$-KG soliton quantization.

5.5 EffectMeasure bridge B1 and B2

Winding-sector Born weights $\mu(P_k) = \sum_s |\alpha_s|^2$ on $\mathbb{Z}_7^5$ satisfy the EffectMeasure axioms for random superpositions and POVM partitions (**A**, 200/200 states tested). The information-loss channel redistributes sector weights but preserves the axiom structure after renormalization. `ThooftEffectMeasureBridge.lean` certifies normalization and nonnegativity (`thooft_b1_master_bundle`, zero `sorry`); full `EffectMeasure` on $\mathrm{Fin}\,7^5$ remains open at the Lean elaboration scale.

The bridge B2 (PSC + f_{MDL} $\Rightarrow$ EffectMeasure) is *derivable*, not axiomatic: B2a (BeableAmplitudeHypothesis $\Rightarrow$ EffectMeasure) is mechanical once the Fock lift supplies amplitudes; B2b (PSC physical state $\Leftrightarrow$ amplitude hypothesis) is the canonical-quantization step (**A/D**). When citing `PSCEffectMeasure.lean`, disclose that the module previously carried seven private kernel-workaround axioms, all of which have been discharged via parametric proofs over `Fin n` in `PSCEffectMeasureGeneric.lean`. The sole remaining imported axiom is `re_trace_psd_mul_psd_nonneg` from `nems-lean`; the module builds with zero `sorry`.

Remark 5.5 (Born rule from MDL-minimality and D5). The present paper treats the Born rule structurally via 't Hooft's information-loss mechanism (**A/D**, Section 5.1). The continuum Φ_{MDL} assignment $P(\mathrm{kink}_k) \propto |\varphi_k|^2$ (D5) is partially certified: Sections 5.4 and 5.5 supply the MDL and Fock constructive layer; full unconditional closure awaits continuum soliton quantization. NEMS

Paper 13 ([3], DOI `10.5281/zenodo.19429741`) remains the PSC uniqueness theorem; the GTE route supplies compatible constructive amplitudes.

5.6 Double-slit Born-rule confirmation

Monte Carlo [**D**] click ensembles on a GTE two-path interference setup reproduce $|\phi|^2$ weights with correlation 0.994 and $\chi^2_{\mathrm{red}} \approx 10^{-4}$ (**A**). Zone L1 Huygens–Fresnel fringes match $\Delta x = \lambda L/d$ to 1.3%. This confirms the structural Born chain at the canonical interference experiment.

5.7 Anomalous magnetic moment of the muon

The GTE one-loop prediction $a_\mu^{\mathrm{GTE}} = \alpha_{\mathrm{GTE}}/(2\pi) = 1/(274\pi) \approx 1.16 \times 10^{-3}$ matches the QED Schwinger term to 0.026% (**A**; inherits P28 [5], one-loop $g-2$ subsection). The framework is **not falsified** at one loop and is **neutral** on the Fermilab anomaly $\Delta a_\mu \approx 2.5 \times 10^{-9}$: two-loop QED gives $a_\mu^{\mathrm{GTE,2L}} \approx 4.13 \times 10^{-6}$ ($\sim 1660\times$ too large); a GTE hadronic HVP dispersion estimate $a_\mu^{\mathrm{HVP,GTE}} \approx 6.2 \times 10^{-8}$ shifts the SM value by -6.4×10^{-9} (wrong sign to explain the anomaly). These assessments are order-of-magnitude only where stated; they do not constitute a GTE prediction of Δa_μ.

6 Lorentz-Invariant Causal Structure

The AFCA architecture used here for causal invariance is the inner-clock gating mechanism of the CMCA (P41); the synchronous f_{MDL} ring used for the Hilbert space construction in Section 2 is its Level-1 algebraic orbit map. These are two roles of one system, not two competing frameworks.

6.1 AFCA causal invariance and Lamport order

A key requirement for Lorentz-invariant physics is that causal relationships between events be independent of the reference frame used to describe them. In the AFCA framework, different Lorentz frames correspond to different choices of update ordering for the asynchronous CA. *Causal invariance* is the property that guarantees frame-independence: it asserts that the causal partial order on events is the same regardless of which valid update schedule is applied. If the underlying CA dynamics are causally invariant, then observers using any Lorentz-compatible update schedule will agree on which events causally precede which others — this is the discrete analog of the Minkowski-causal-cone structure being Lorentz-invariant.

The AFCA (Asynchronous Fractal Cellular Automata; [7]) architecture, whose inner first-passage time τ_c governs the fractal cell-gating mechanism, defines a causal graph whose edges depend on lattice position and update ordering, not on cell colour content. The graph is *causally invariant*: every valid update schedule yields the same Lamport partial order on events. In Lean 4 this is machine-certified as `lamport_strict_partial_order` in `CausalInvariance.lean` (zero `sorry`): the AFCA causal relation is a strict partial order independent of the synchronous outer-step gauge. This is a *derived* property of the GTE CA dynamics, not a postulate: it follows from the local structure of the AFCA causal graph together with the f_{MDL} rule, and is proved without any Lorentz-invariance assumption as input.

Chiral glider trajectories on the two-layer Rule 110/Rule 124 pair satisfy a discrete light-cone bound with effective signal speed $c_{\mathrm{eff}} = 2/3$ in lattice units (`chiral_trajectory_light_cone`, $\mathbf{A}_{\mathrm{Lean}}$). The inclusion of Minkowski causal cones into this discrete structure is certified by `minkowski_` `causal_isomorphism` ($\mathbf{A}_{\text{Lean}}$): every future-directed Minkowski interval with $|\Delta x| \le (2/3)\,\Delta t$ has a preimage in the AFCA causal graph.

6.2 Special-relativistic time dilation

The Lamport order supplies the causal scaffolding for Lorentz-invariant physics: macroscopic observers reading the AFCA clock field τ_c measure proper-time ratios that scale with the Lorentz factor $\gamma(v)$ for moving glider patterns (**A**). Definitive paired comparisons give 10/15 velocity pairs within 15% of $\gamma(v)$, with best agreement 8.7% at $\gamma = 1.658$ and mean error 12.0% at the single-level ($M = 7, N = 1$) resolution used in the present tests [7]. The residual is consistent with expected finite-lattice corrections that vanish in the continuum limit (**D**: full Lean 4 closure of SR awaits the bijection results below). The DWeight-weighted [**D**]-average of τ_c tightens this to 5.4% raw error (1.2–1.8% after lattice correction; 0.069% mean error across $\beta \in [0.05, 0.90]$ in the Φ_{MDL} continuum): only PSC-admissible (canonical GTE A-glider) beables satisfy the SR relation, while non-canonical patterns deviate by 54–105%, confirming the physical selectivity of the DWeight projection — machine-certified with zero `sorry` (`dmdl_qec_sr_bundle`, `DWeightSRFormula.lean`, $\mathbf{A}_{\text{Lean}}$).

6.3 Coordinate surjection in the continuum limit

The Minkowski–AFCA correspondence is strengthened by the continuum-limit coordinate surjection `minkowski_causal_surjection_continuum_limit` ($\mathbf{A}_{\text{Lean}}$, `CausalInvariance.lean` §4, zero `sorry`): for every lattice point $(t, x) \in \mathbb{N}^2$ in the Minkowski coordinate chart, there exists an AFCA causal-graph event at sufficiently large tape size (L, T). Inclusion of discrete causal events into the Minkowski cone is therefore closed at the level of coordinate surjection.

Remark 6.1 (Honest scope: bijection open). The coordinate surjection does *not* establish a bijection between Minkowski events and AFCA causal nodes: multiple graph events may map to the same continuum coordinate, and the reverse map from continuum coordinates to unique graph nodes remains **D** open. Full Lorentz-invariant CA→QFT embedding therefore requires the bijection layer in addition to the surjection and light-cone inclusion already certified.

The SR mechanism on the AFCA substrate — clock-field definition, chiral-pair light cone, and τ_c ratio tests — is developed in [7]; the present section records the causal-structure theorems that connect that emergent-spacetime programme to Lorentz-invariant quantum mechanics on the GTE substrate.

7 Open Problems

The results of this paper establish a coherent quantum mechanical structure for the GTE CA at the **A/A/D** level. The following gaps remain open.

7.1 Continuum limit and Schrödinger equation

The most significant open problem is the continuum limit: passing from the discrete $\mathbb{Z}_7^5$ CA to a continuous quantum field theory with a Schrödinger or Dirac equation. This is the lattice-to-continuum problem, analogous to recovering continuum QFT from lattice QCD. Two components should be distinguished: (a) the *algebraic* content of the Φ_{MDL} continuum limit — the unique terminal Level-1 GTE object with Nyquist residual $\varepsilon_0(M) = \pi^2/(3M^2) \to 0$ as $M \to \infty$ — is $\mathbf{A}_{\text{Lean}}$

conditional in P41/P42 [9, 13] (`cmca_continuum_limit_is_phimdl`, `CMCAContinuumLimit.lean`, zero `sorry`); (b) the full Minkowski bijection (smooth $\mathrm{SO}(1,3)$ recovery) remains **D** open.

The discrete Schrödinger evolution $U(n) = e^{-iH_{\mathrm{cyc}}\, n\tau_c}$ of the CMCA cogwheel states on $\mathbb{Z}_7^5$ is Lean-certified with zero sorry: the composition law $U(m+n) = U(m)\, U(n)$, unitarity, and the step recurrence $U(n+1) = e^{-iH_{\mathrm{cyc}}\tau_c}\, U(n)$ all hold for Hermitian H_{cyc} (`CogwheelDynamicsG21.lean`, `UgpLean.QuantumDynamics.G21`, $\mathbf{A}_{\mathrm{Lean}}$; master bundle `g21_quantum_dynamics_catad`, zero `sorry`). The Level-2 Klein-Gordon → chiral-Schrödinger factorisation at the $(+,-)$-doublet level — $(i\partial_t - \omega_k)\varphi_k = 0$ per mode — is a **A/D** structural observation independent of the Level 1 → Level 2 continuum limit. Full continuous-time $\psi(t) = e^{-iHt}\psi(0)$ at Level 2 requires the G22/G42 inductive-limit closure.

No claim to a continuum derivation is made in the present paper. The 1-dimensional Hilbert space $\mathcal{H}_{\mathrm{phys}} = \mathbb{C}|\mathrm{vac}\rangle$ describes only the ground state of the visible sector; a full multi-particle Hilbert space requires specifying a tensor-product structure over many copies or cells of the CA.

7.2 Lean certification of the gauge identification

Theorem 4.6 is currently **A/D**. The arithmetic components (the $\mathbb{Z}_7 \to Q$ charge map, the $Z_3 \subset \mathbb{Z}_7^*$ subgroup, and the $\{W=3, W=4\}$ doublet structure) are each $\mathbf{A}_{\mathrm{Lean}}$ (Lean-certified with zero sorry). The remaining gap is the identification of the winding-equivalence classes as 't Hooft's information-equivalence gauge classes in the formal sense of §9.3 of [15]. Closing this gap — formally stating and certifying the 't Hooft §9.3 identification for the GTE CA in Lean 4 — would upgrade the entire gauge derivation to $\mathbf{A}_{\mathrm{Lean}}$.

7.3 Multi-particle Hilbert space

The present paper analyses the single-ring $\mathbb{Z}_7^5$ visible sector. A physical multi-particle Hilbert space requires a tensor-product construction over multiple rings.

The algebraic layer of that construction is formalized in Lean 4 (`MultiParticleHilbert.lean`, zero `sorry`, $\mathbf{A}_{\mathrm{Lean}}$). The key certified results are:

- **Code word count ($\mathbf{A}_{\mathrm{Lean}}$).** There are exactly 4 GTE code words $\{\mathrm{vac}, \mathrm{gen}_1, \mathrm{gen}_2, \mathrm{gen}_3\} \subset \mathbb{Z}_7^5$ (`code_word_cardinality`, certified via an explicit bijection with `Fin 4`).
- **N-particle dimension ($\mathbf{A}_{\mathrm{Lean}}$).** The N-particle state space `NParticleState n` $= \mathrm{Fin}(N) \to$ CodeWord has Fintype cardinality 4^N (`n_particle_state_count`).
- **Normalized DWeight ($\mathbf{A}_{\mathrm{Lean}}$).** The DWeight product of any multi-beable state equals 1 (`multiDWeight_eq_one`), reflecting that every code word is a unit-weight physical state.
- **Additive mass observable ($\mathbf{A}_{\mathrm{Lean}}$).** The total GTE orbit-index mass of an N-particle state is additive (`multiMass_append`) and bounded by $3N$ (`multiMass_le`). The spectrum is $\{0,1,2,3\}^{\times N}$.
- **Physical mass anchor ($\mathbf{A}_{\mathrm{Lean}}$).** Every non-vacuum code word carries physical mass ≥ 1.8 MeV (`smGenMass_multi_anchor`), anchoring the abstract orbit-index mass to the PDG mass scale. Since all nine SM fermion masses are reproduced by the GTE IMT without free parameters (**A**), the multi-particle mass operator has a fully determined spectrum.
- **Stabilizer closure ($\mathbf{A}_{\mathrm{Lean}}$).** The f_{MDL} stabilizer maps every code word in a multi-state to a code word (`multiparticle_orbit_closure`), so the code subspace is preserved under particle evolution.

- **Orthonormal basis ($\mathbf{A}_{\text{Lean}}$).** The Kronecker basis inner product on the N-particle state space is positive definite: $\langle\psi|\psi\rangle = 1 > 0$ for all basis states (`inner_product_positive_definite`).

The appropriate tensor-product structure for the GTE CA in the 't Hooft information-equivalence sense (connecting the code-word basis to the cogwheel construction of §2) remains open (**D**). The full Hilbert space completion (ℓ^2 norm, continuum limit) requires the AFCA architecture (open problem OQ-CL1).

7.4 CMCA Hilbert space to Φ_{MDL} Fock space bridge

The sector embedding maps from the CMCA Hilbert space to the Φ_{MDL} Fock space [9] are Lean-certified (zero `sorry`, `CMCAHilbertFockBridge.lean`, `ugp-lean`):

- **Vacuum map ($\mathbf{A}_{\text{Lean}}$).** The zero-kink Fock vacuum corresponds to the CMCA winding-0 sector (`fock_vacuum_maps_to_cmca_vacuum`).
- **1-particle sector maps ($\mathbf{A}_{\text{Lean}}$).** Each GTE kink mode embeds as a canonical 1-particle Fock state (`gTe_kink_mode_maps_to_fock_1particle`).
- **Sector Born rule ($\mathbf{A}_{\text{Lean}}$).** Sector Born weights equal $|c_k|^2$ on the certified beable partition (`born_rule_bridge_from_fock_lift`).
- **PSC admissibility ($\mathbf{A}_{\text{Lean}}$).** GTE kink-mode windings lie in the PSC-admissible sector set $\{0, 2, 3, 4, 6\}$ (`kink_mode_windings_psc_admissible`).
- **Sector count and beable lift ($\mathbf{A}_{\text{Lean}}$).** Five PSC sectors and four GTE-labelled kink Fock modes are certified (`cmca_fock_sector_count`, `bps_psc_sector_has_beable_lift`).

G42 is **A/D**: the G22 sector embedding maps are fully $\mathbf{A}_{\text{Lean}}$ (12 zero-`sorry` theorems, `CMCAHilbertFockBridge.lean`), and the CMCA continuum limit (`cmca_continuum_limit_is_phimdl`, `CMCAContinuumLimit.lean`) is $\mathbf{A}_{\text{Lean}}$ zero-`sorry`. Both conditions are $\mathbf{A}_{\text{Lean}}$, closing the conditional decomposition (`ca_qft_embedding_reduces_to_g22`). The remaining **D** gap — the full analytic Jackiw–Rebbi + GNS construction embedding $H_{\text{phys}}(L) \hookrightarrow H_{\text{Fock}}$ dense as $L \to \infty$ — requires a Lean GNS/operator-topology library not yet available, placing the overall G42 confidence at **A/D**.

7.5 Quantitative mass hierarchy from tail lengths

The tail-length ordering $3 > 2 > 1$ matches the stability hierarchy qualitatively, and this ordering is machine-certified in Lean 4 (`tail_length_strict_ordering` and `mass_ordering_from_tail_length`, GUTStructure §76, zero `sorry`, $\mathbf{A}_{\text{Lean}}$): gen_1 (tail = 3) is the lightest and gen_3 (tail = 1) is the heaviest.

The N_{eff} values $\{73, 42, 275\}$ are not monotone in tail length: $b_{\text{gen}_3} = 275 \gg b_{\text{gen}_1} = 73$ despite $\text{tail}(\text{gen}_3) = 1 < \text{tail}(\text{gen}_1) = 3$ (machine-certified: `neff_not_monotone_in_tail`, GUTStructure §76, zero `sorry`, $\mathbf{A}_{\text{Lean}}$). The quantitative lepton mass ratios $m_\mu/m_e = N_{\text{eff}}(\mu)/N_{\text{eff}}(e) = 42/73$ and $m_\tau/m_e = N_{\text{eff}}(\tau)/N_{\text{eff}}(e) = 275/73$ require the GTE cascade formula of P02 and are not derivable from tail lengths alone. The best-fit polynomial $b(n) = 772 - 629n + 132n^2$ interpolates the three data points exactly, but its coefficients ($772 = 2^2 \cdot 193$, $629 = 17 \cdot 37$, $132 = 2^2 \cdot 3 \cdot 11$) have no GTE-structural interpretation; a three-parameter fit to three points is trivially exact and does not constitute a derivation. The cascade seed sum $b_{\text{gen}_1} + b_{\text{gen}_2} + b_{\text{gen}_3} = 390 = 2 \cdot 3 \cdot 5 \cdot 13$ is machine-certified (`mass_quantitative_formula_requires_cascade`, GUTStructure §76, zero `sorry`, $\mathbf{A}_{\text{Lean}}$); the quantitative ratios follow from the N_{eff} cascade (open problem from tail lengths alone). The structural near-miss (75 tail-3 states vs. $N_{\text{eff}}(\text{gen}_1) = 73$, offset $= 2$, certified `tail_neff_nearness`) is interesting but does not constitute a derivation.

7.6 Multiway causal graph formal map

The Wolfram Physics Project derives quantum mechanics from the structure of the multiway causal graph of an abstract rule-application system [16] (§§8.12–8.14). Several mechanisms in that framework are structurally parallel to results in the present paper (§§5.1–5.3): the information-loss grounding of quantum probability, the Born-rule branch-weight interpretation, and the measurement-as-branch-selection picture.

A formal map between the Wolfram multiway graph foliation and the Zone L2 / transputation framework of the GTE CA would, if established, provide an independent derivation of the quantum measurement framework from CA-level dynamics. The structural obstacle is that f_{MDL} uses synchronous update — all five cells evolve simultaneously — so the multiway causal graph of f_{MDL} is trivially single-branched: no asynchronous application ordering exists. Establishing a non-trivial multiway structure for the GTE CA, or identifying the appropriate generalisation of the Wolfram framework to synchronous CAs, is an open problem (**D**).

Summary

The 't Hooft cogwheel construction on the GTE f_{MDL} CA on $\mathbb{Z}_7^5$ yields the following results:

1. **Hilbert space (A):** One-dimensional, $\mathcal{H}_{\text{phys}} = \mathbb{C}|\text{vac}\rangle$, $\hat{H}|\text{vac}\rangle = 0$, spectral gap 2π.

2. **Information-loss regime (A):** f_{MDL} is a Chapter 7 (information-loss) automaton: 98.71% GoE states, universal vacuum attractor in ≤ 7 steps.

3. **Stability hierarchy (A):** Tail-length ordering $\ell(\text{gen}_1) > \ell(\text{gen}_2) > \ell(\text{gen}_3)$ matches the empirical generation stability hierarchy.

4. **Eigenvalue-mass falsification (A):** Direct eigenvalue-to-mass correspondence fails on all formulations tested.

5. **Two-Role Structural Principle (A/D):** Cogwheel provides QM structure (Lean-certified); GTE arithmetic provides mass content. These are non-redundant complementary contributions; the inter-layer identification is a conditional derivation (organizing principle, not a machine-certified theorem).

6. **Gauge derivation (A/D, $\mathbf{A}_{\text{Lean}}$ components):** G_{SM} from $\mathbb{Z}_7$ winding-equivalence classes via 't Hooft's information-equivalence identification. $U(1)_{\text{EM}}$, $SU(2)_L$, $SU(3)_c$, and $U(1)_Y$ hypercharge consistency (including Weinberg angle $\sin^2\theta_W = 3/13$) are all $\mathbf{A}_{\text{Lean}}$. Sector $W = 1$ predicts the SU(5) Y-leptoquark mediator ($Q = +\frac{1}{3}$); the X-leptoquark ($Q = +\frac{4}{3}$) cohabits $W = 4$ with the electron.

7. **Dark sector ($\mathbf{A}_{\text{Lean}}$):** Five-dimensional dark Hilbert space ($E = 0$ plus $E = \pi$ doublets), structurally explaining dark-sector particle stability.

8. **Born rule (A/D):** Structural derivation from information loss; compared to independent UGP derivation. MDL derivation of $P(kink) \propto |\varphi|^2$ from [D] remains open (Theorem 5.5).

9. **Lorentz causal structure ($\mathbf{A}_{\text{Lean}}$/A):** Lamport causal invariance, Minkowski light-cone inclusion, and coordinate surjection certified; full causal-node bijection **D** open (Section 6).

10. **Multi-particle state space ($\mathbf{A}_{\text{Lean}}$):** The algebraic layer of the multi-particle Hilbert space is Lean 4-certified (`MultiParticleHilbert.lean`, zero `sorry`): 4 code words, 4^N basis states for N particles, DWeight product $= 1$, additive mass observable with spectrum $\{0, \ldots, 3N\}$, physical mass floor ≥ 1.8 MeV per non-vacuum particle, and a positive-definite Kronecker basis inner product. The full Hilbert space completion (continuum limit) remains **D**.

The principal open gaps are the continuum limit (**D**) and the Minkowski–AFCA bijection (Theorem 6.1).

A Lean Certification Inventory

Table 5 lists machine-certified theorems cited in this paper. All modules are in the canonical `ugp-lean` repository (zero `sorry`, $\mathbf{A}_{\text{Lean}}$, unless noted).

Theorem	**Module**	**Role**
`su2l_charge_assignment_z7_discriminator`	`GUTStructure`	$\{W{=}3, W{=}4\}$ doublet arithmetic
`phimdl_potential_su2l_invariant`	`GaugeMDL.lean`	L_2 preserved under $SU(2)_L$
`su2l_covariant_derivative_minimal`	`GaugeMDL.lean`	MDL-minimal covariant derivative
`su2l_wpm_generator_algebra`	`GaugeMDL.lean`	$W^{\pm}$ Lie algebra
`su2l_l2_from_phimdl_potential_catad`	`GaugeMDL.lean`	$SU(2)_L$ from PMDL; zero named axioms
`multiparticle_space_well_defined`	`MultiParticleHilbert.lean`	N-particle code-word Hilbert layer

Table 5: Lean 4 certification inventory (selected theorems).

Acknowledgments

This work is part of the UGP Physics programme, the quantitative branch of the Reflexive Reality programme [12].

AI coding assistants were used for LaTeX formatting, coding assistance, and adversarial review during manuscript preparation. All mathematical derivations, numerical computations, and Lean 4 certifications were independently verified by deterministic scripts and the Lean 4 kernel.

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